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Enhancing Robustness: Delay-Based Control and Observer Strategies for Tilt-Rotor Quadrotors*

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***Abstract*— This paper presents a new control scheme, named the Proportional Delayed Integral (PδI) controller, for achieving tracking tasks on a tilt-rotor quadrotor. The proposed approach integrates a nonlinear disturbance observer, which effectively mitigates adverse effects stemming from parametric inaccuracies or unknown forces. By enhancing the traditional quadrotor with a tilting mechanism, the operational range is extended, enabling interaction with the surroundings via a robotic arm with one degree of freedom. To ensure precision and adaptability during operations, a comprehensive mathematical model has been meticulously developed, accurately reflecting the unique capabilities of the drone. The synergistic combination of the PδI controller with the nonlinear disturbance observer empowers the tilt-rotor quadrotor to tackle complex tasks with efficiency and reliability. Through rigorous numerical simulations demonstrating the effectiveness of this design, this study lays the foundation for a new era of highly functional and versatile drones, poised to revolutionize various industries and applications.**

I. Introduction

Unmanned aerial vehicles (UAVs) have come a long way, with quadrotors emerging as a leading design. These versatile and maneuverable aircraft, characterized by their four rotors in a square configuration, have become the most popular UAVs due to their simple design and ease of control. Each rotor generates lift and thrust, allowing quadrotors to hover, move in all directions, and perform vertical takeoff and landing (VTOL) without a runway. This unique configuration grants them exceptional agility, making them ideal for a vast array of tasks and environments.

One promising application for quadrotors is urban air mobility (UAM) [1] and cargo transportation. As cities grapple with traffic congestion and pollution, flying taxis offer a futuristic solution. Quadrotors, with their VTOL capabilities and compact size, could navigate densely populated areas, providing on-demand transportation and potentially alleviating traffic woes and environmental concerns.

Furthermore, quadrotors excel at carrying payloads, ranging from small packages to medical supplies. Their ability to access remote locations makes them invaluable for delivering essential goods during natural disasters or emergencies [2]. Advancements in payload capacity and flight range hold the potential to revolutionize logistics and supply chain management [3].

The impact of quadrotors extends beyond transportation. In agriculture [4], they are used for crop monitoring, spraying pesticides, and precision agriculture, improving yields and reducing resource waste. Construction and infrastructure benefit from their ability to survey sites, monitor progress, and inspect structures [5]. Filmmaking and entertainment leverage quadrotors to capture breathtaking aerial footage, enhancing storytelling. Search and rescue operations, surveillance and security, environmental monitoring, and scientific research all utilize quadrotors due to their versatility and maneuverability. They offer valuable tools in situations where traditional methods are impractical or inaccessible.

As quadrotors become more sophisticated, advanced control approaches are essential for stable and precise flight, especially in complex environments. Nonlinear control techniques [6], such as model predictive control (MPC) [7] and sliding mode control (SMC) [8], offer robustness and adaptability to varying conditions and disturbances.

Delay-based controllers [9]–[15] specifically address challenges associated with communication delays and sensor latency in UAV systems [16]. By incorporating predictive algorithms and feedback mechanisms, they enhance responsiveness and reliability, enabling effective operation in dynamic environments.

Despite their potential, several challenges remain before quadrotors become mainstream: regulatory hurdles, airspace integration, safety concerns, and public acceptance. Collaboration between governments, industry stakeholders, and developers is crucial for safe and efficient operation in urban airspaces.

Technical challenges such as limited flight endurance, payload capacity, and noise emissions also pose significant barriers. Advancements in battery technology, propulsion systems, and autonomous navigation algorithms are necessary to overcome these limitations and unlock the full potential of quadrotors in future mobility solutions.

Observing and mitigating disturbances are essential aspects of quadrotor control systems, particularly in dynamic and uncertain environments. Disturbance observers [17] and adaptive control strategies [18] enable quadrotors to detect and counteract external factors like wind gusts, turbulence, or payload variations, ensuring stable flight performance.

This paper enhances quadrotor control by combining a PδI controller with disturbance observation techniques, im-

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proving both tracking accuracy and system resilience against perturbations. Unlike prior studies that separately focused on PID controllers and Linear Quadratic Regulators, this integrated approach offers a more robust solution.

II. Mathematical model

In this work, we adopt a non-holonomic flight approach to streamline the navigation task of an interactive tilt-rotor quadcopter. Our primary goal is to achieve smooth and precise movements, particularly to facilitate the manipulation of its one-degree-of-freedom robotic arm. We focus on a $2-D$ analysis within the $h-z$ axis framework, as depicted in Figure 1, emphasizing altitude and heading movements.

Consequently, our paper addresses the tracking task within the Q−plane, considering the non-holonomic constraints governing motion along this plane.

A. Generalized coordinates and control inputs

Generalized coordinates are a minimal set of parameters used to describe the kinematics of a tilt-rotor quadcopter during tracking tasks in the Q−plane. Let us define the generalized coordinates as $\mathbf{q} := \begin{bmatrix} h & z & \theta & \gamma \end{bmatrix}^T$, where h and z represent lateral and vertical movements, respectively, and θ and γ, denote the tilt of the aircraft and the motion of the robotic arm, respectively.

For the fully actuated tilt-rotor quadcopter, the control inputs are $\boldsymbol{u} := \begin{bmatrix} u_h & u_z & u_\theta & u_\gamma \end{bmatrix}^T$, influencing the quadcopter's flight. These inputs relate to the thrust of each rotor as follows:

$$\begin{bmatrix} u_h \\ u_z \\ u_\theta \\ u_\gamma \end{bmatrix} = \begin{bmatrix} T\sin(\theta+\beta) \\ T\cos(\theta+\beta) \\ (T_1 - T_2 + T_3 - T_4)\ell_r \cos(\theta+\beta) \\ f_p l_p \end{bmatrix},$$

where β is the rotor tilt angle, ℓ_r is the distance from the center of gravity to each rotor, f_p is the force from the robotic arm, l_p is the distance from the center of gravity to the arm's force application point, and T is the total thrust from all rotors ($T = T_1 + T_2 + T_3 + T_4$).

B. Dynamical model via Euler-Lagrange approach

The tilting-quadrotor configuration, as illustrated in Fig. 1, reveals two interconnected subsystems. The following analysis considers the additive property of the Lagrangian to obtain a dynamical model examining both subsystems separately. Firstly, the analysis focuses on the kinetic and potential energy of the quadrotor's subsystem and then shifts to the robotic arm system, conceptualized as a pendulum. This analysis leads to the Lagrangian:

$$\begin{aligned}\mathcal{L} = &\frac{1}{2}(m_r+m_p)\left(\dot{h}^2+\dot{z}^2\right)+m_p l_p(\dot{\theta}+\dot{\gamma})\left(\cos(\theta+\gamma)\dot{h}+\sin(\theta+\gamma)\dot{z}\right) \\ &+\frac{1}{2}I_r\dot{\theta}^2+\frac{1}{2}\mathrm{I}_p(\dot{\theta}+\dot{\gamma})^2-(m_r+m_p)gz+m_p g l_p\cos(\theta),\end{aligned} \tag{1}$$

where m_r, m_p are the masses of the quadrotor and pendulum, θ is the rotation of the drone, γ angular position of the pendulum, $\mathrm{I_r}$, $\mathrm{I_p}$ denote the moments of inertia of the quadrotor and pendulum, l_p represents the length of the robotic arm and g denotes the acceleration of gravity. For further details, please review [19].

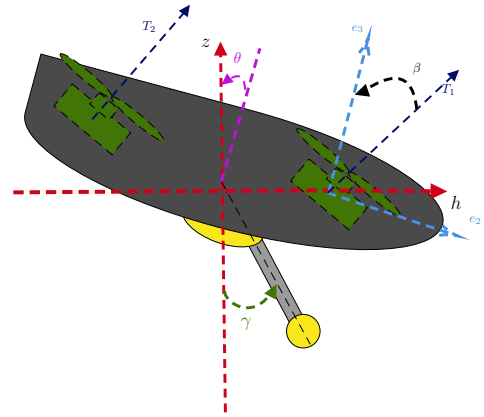

Fig. 1. Forces exerted on tilt-rotor UAV in the $Q-$plane.

To obtain the dynamic motion of the system, we utilize the Euler-Lagrange (E-L) approach, resulting in:

$$M(\boldsymbol{q})\ddot{\boldsymbol{q}} + C(\dot{\boldsymbol{q}},\boldsymbol{q})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{u}, \tag{2}$$

where the matrices $M(\boldsymbol{q})$, $C(\dot{\boldsymbol{q}},\boldsymbol{q}) \in \mathbb{R}^{4\times4}$ and $\boldsymbol{g}(\boldsymbol{q}) \in \mathbb{R}^4$ are given as follows:

$$M(\boldsymbol{q}) := \begin{bmatrix} m_r+m_p & 0 & m_a & m_a \\ 0 & m_r+m_p & m_b & m_b \\ m_a & m_b & \mathrm{I}_r+\mathrm{I}_p & \mathrm{I}_p \\ m_a & m_b & \mathrm{I}_p & \mathrm{I}_p \end{bmatrix}$$

$$C(\dot{\boldsymbol{q}},\boldsymbol{q}) := \begin{bmatrix} 0 & 0 & c_c & c_c \\ 0 & 0 & c_d & c_d \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \ \boldsymbol{g}(\boldsymbol{q}) := \begin{bmatrix} 0 \\ (m_r+m_p)g \\ m_p l_p g\sin(\theta) \\ m_p l_p g\sin(\theta) \end{bmatrix},$$

whith $m_a := m_p l_p\cos(\theta+\gamma)$, $m_b := m_p l_p\sin(\theta+\gamma)$, $c_c := -m_p l_p\sin(\theta+\gamma)(\dot{\theta}+\dot{\gamma})$, $c_d := m_p l_p\cos(\theta+\gamma)(\dot{\theta}+\dot{\gamma})$.

III. Control Strategy

This paper addresses trajectory tracking challenges in tilt-rotor quadrotors. We propose a delay-based control approach that ensures precise movement and minimal deviations. Delay-based controllers offer the advantage of including an additional degree-of-freedom parameter that can be adjusted to improve performance. Building on prior work [20], we reformulate the quadrotor dynamics (2) using a change of variables. This new system, denoted by $\widetilde{\boldsymbol{q}} := \boldsymbol{q} - \boldsymbol{q}_d$ where $\boldsymbol{q}_d$ is a smooth reference.

By exploiting this transformed system and applying principles of nonlinear control, we develop a control law that guarantees both stability and precise trajectory tracking. The control input for this law is expressed as:

$$\boldsymbol{u}(\dot{\boldsymbol{q}},\boldsymbol{q}) = M(\boldsymbol{q})[\ddot{\boldsymbol{q}}_d - K_p\widetilde{\boldsymbol{q}} - K_i\widetilde{\boldsymbol{q}}_\tau] + C(\dot{\boldsymbol{q}},\boldsymbol{q})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}), \tag{4}$$

where K_p, $K_i \in \mathbb{R}^{4\times4}$ are the proportional and integral gains of the PδI controller [15] and $\widetilde{\boldsymbol{q}}_\tau := \int_0^t \widetilde{\boldsymbol{q}}(v-\tau)dv$. Firstly, to tune the P$\delta$I controller, it must be ensured that the system is stable in the closed loop, so the following result is presented.

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Proposition 1: Let $\boldsymbol{q}_d$ be a suitably smooth desired trajectory and τ a fixed delay. Consider the system (2) in closed-loop with control (4), where K_p and K_i are real diagonal matrices defined as $K_p := \mathrm{diag}\{k_{p_1}, \cdots, k_{p_4}\}$, $K_i := \mathrm{diag}\{k_{i_1}, \cdots, k_{i_4}\}$. For $j \in \{1, \ldots, 4\}$ and $\ell \in \mathbb{N}$, assume that the gains k_{p_j} and k_{i_j} are chosen by satisfying the following inequalities:

$$\tau > 0, \tag{5}$$

$$\left(\frac{(2\ell-1)\pi}{2\tau}\right)^2 < k_{p_j} < \left(\frac{(2\ell+1)\pi}{2\tau}\right)^2, \tag{6}$$

$$0 < k_{i_j} < k_{p_j}\left(\frac{\pi}{2\tau}\right) - \left(\frac{\pi}{2\tau}\right)^3. \tag{7}$$

Finally, the closed-loop system is asymptotically stable.

Remark 1: According to condition (5) stated in Proposition 1, it is evident that the system requires a positive delay, meaning that it cannot be stabilized using a classical PI controller, i.e., the polynomial $\Delta(s; k_p, k_i) := s^3 + k_p s + k_i$ is not stabilizable, since it does not satisfy the Stodola condition [21].

On the other hand, the controller can also be tuned using the $\mathcal{D}$-Decomposition methodology which generates stability maps in terms of the design parameters (k_p, k_i, τ).

Proposition 2: Let $h > 0$ be a fixed value, the set of all stability crossing curves is composed by the following set of lines:

$$k_i := 0, \quad k_p \in \mathbb{R}. \tag{8}$$

$$k_i := (-1)^\ell \left(\omega_\ell^3 - k_p \omega_\ell\right), \quad k_p \in \mathbb{R}. \tag{9}$$

where $\omega_\ell = \dfrac{(2\ell-1)\pi}{2\tau}$, and $\ell \in \mathbb{N}$.

Fig. 2 depicts the behavior of the delay-based controller parameters as τ varies from 0.5s to 1.5s. In particular, we see that the stability zones collapse as the delay rises, resulting in a smaller range of values from which to choose the controller gains. As a result, the most crucial element to balance is the delay, as a short delay maximizes the value of the benefits.

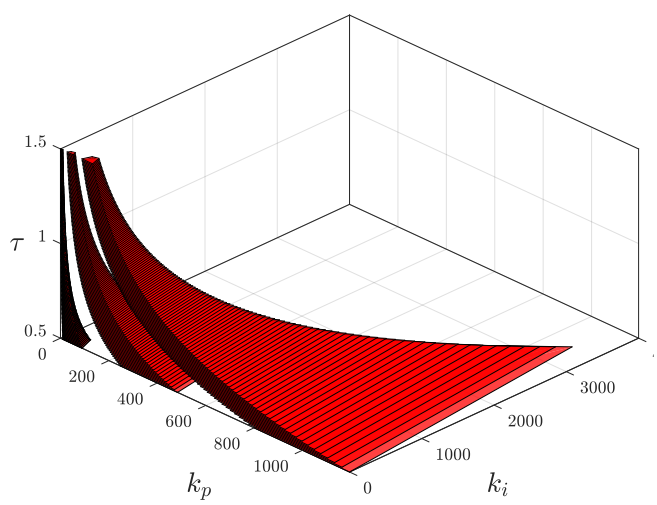

Fig. 2. Stability regions (k_p, k_i, τ) where $\ell = 1, 2, \ldots, 6$.

In this sense, one of the main tools to design a controller that guarantees such a convergence relation is the notion of $\sigma-$stability. To state such a notion, let be $\sigma \in \mathbb{R}_+$. Then, the stability-problem can be described as the task of determining the gains of the controller $\boldsymbol{k}_j := (k_{pj}, k_{ij})$ for which the real part of the rightmost roots of the characteristic function for the closed-loop system is smaller than $-\sigma$.

Proposition 3: Let $\tau > 0$ be a fixed value, $\sigma > 0$, the set of all stability crossing curves is composed by

$$k_p(\omega, \sigma) = 3\omega^2 - \sigma^2 + \frac{2\omega(\omega^2 + \sigma^2)\cos(\omega h)}{\sigma \sin(\omega h) - \omega \cos(\omega h)}, \tag{10}$$

$$k_i(\omega, \sigma) = e^{-h\sigma} \frac{2\omega(\omega^2 + \sigma^2)\sigma}{\sigma \sin(\omega h) - \omega \cos(\omega h)}. \tag{11}$$

Sketch of proof: The proof is performed in the same way as for Proposition 2 with $s = -\sigma + \boldsymbol{i}\omega$. ■

By choosing a delay value, the regions of stability are delimited by the stability crossover curves shown in Fig. 3, where a range of possible combinations of k_{pj} and k_{ij} values are represented within the yellow zone, guaranteeing the stability of the system.

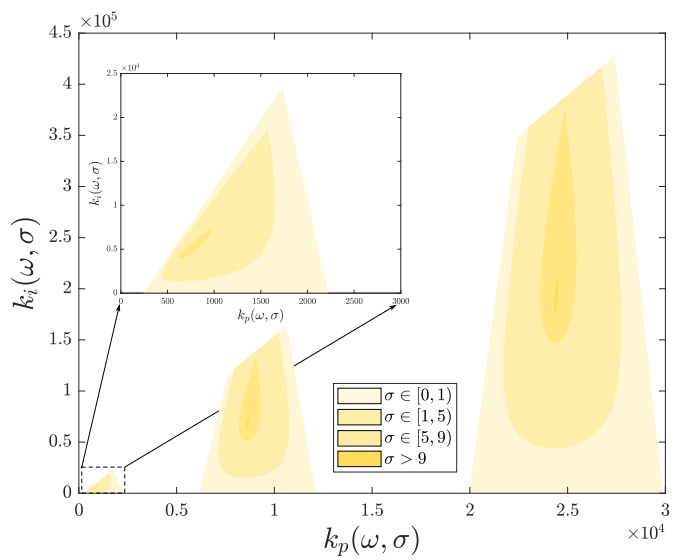

Fig. 3. Stability regions (k_p, k_i) where $0 < \sigma < 9$ and $\tau = 0.1$.

IV. Nonlinear Disturbance Observer (NDO)

This work integrates an aerial robot with a robotic arm to interact with its environment. The robotic arm's payload manipulation generates disturbances due to parametric inaccuracies and external factors. To consider the disturbances, the dynamical model is expanded to include uncertainties, adjusting the equation to:

$$M(\boldsymbol{q})\ddot{\boldsymbol{q}} + C(\dot{\boldsymbol{q}}, \boldsymbol{q})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{u} + \boldsymbol{d}, \tag{12}$$

where $\boldsymbol{d}$ represents disturbances from unknown parameters and external forces. The observation error, defined as the difference between the measured and observed disturbances, is:

$$\boldsymbol{e_o} := \boldsymbol{d} - \widehat{\boldsymbol{d}}. \tag{13}$$

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Without prior knowledge of $\boldsymbol{d}$, it is commonly assumed that:

$$\dot{\boldsymbol{d}} \equiv \boldsymbol{0}. \tag{14}$$

This assumption implies that disturbances change slowly compared to the observer's adaptation ability, allowing the observer to keep up with variations. Assuming that (14) holds, we need error dynamics that ensure the observation error eventually goes to zero. Similar to classical nonlinear observers, we propose the following error dynamics:

$$\dot{\boldsymbol{e}}_{\boldsymbol{o}} = -L(\boldsymbol{q}, \dot{\boldsymbol{q}})\boldsymbol{e}_{\boldsymbol{o}}. \tag{15}$$

It is important to consider that $L(q, \dot{\boldsymbol{q}})$ matrix must be positive definite and designed according to the E-L system, and K_o is a diagonal gain matrix. With well-designed error dynamics, a controller can effectively reduce disturbances.

We will use a nonlinear disturbance observer designed for manipulator robots [22], adapted for rotorcraft disturbances. Consider the variable change:

$$\boldsymbol{z}_o := \widehat{\boldsymbol{d}} - \boldsymbol{p}(\boldsymbol{q}, \dot{\boldsymbol{q}}), \tag{16}$$

where $\boldsymbol{p}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ satisfies:

$$\dot{\boldsymbol{p}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = L(\boldsymbol{q}, \dot{\boldsymbol{q}})M(\boldsymbol{q})\ddot{\boldsymbol{q}}. \tag{17}$$

By analyzing the combined dynamics of (16) with (17) and the observer dynamics (15) we can ensure the asymptotic stability of the nonlinear disturbance observer.

Remark 2: Equation (17) offers a key advantage: it eliminates the need to directly measure the system's acceleration $\ddot{\boldsymbol{q}}$. This translates to fewer sensors required for disturbance estimation, simplifying the overall system design.
Thus, the NDO will be considered as:

$$\widehat{\boldsymbol{d}} = \boldsymbol{z}_o + \boldsymbol{p}(\boldsymbol{q}, \dot{\boldsymbol{q}}), \tag{18}$$

where $\boldsymbol{z}_o \in \mathbb{R}^4$ is the change of variable with time derivative given by

$$\dot{\boldsymbol{z}}_o = -L(\boldsymbol{q}, \dot{\boldsymbol{q}})(\boldsymbol{z}_o + \boldsymbol{p}(\boldsymbol{q}, \dot{\boldsymbol{q}}) - C(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{u}). \tag{19}$$

Finally yet importantly, $\boldsymbol{p}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^4$ is the designed function vector that ensures the positive definition of $L(\boldsymbol{q}, \dot{\boldsymbol{q}})$. Considering this designed function vector as

$$\boldsymbol{p}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = K_o^{-1}\left[\dot{h}\ \dot{z}\ \dot{\theta}\ \dot{\gamma}\right]^T, \tag{20}$$

We can define K_o as a diagonal matrix with elements on the diagonal denoted by k_{o1}, k_{o2}, k_{o3} and k_{o4}. To achieve good observer performance, we need to carefully select these gains satisfying the following conditions:

(i) $k_{oj} \in \mathbb{R}_+$, $j \in \{1, \ldots, 4\}$,

(ii) $\max\{k_{o1}, k_{o2}\} < k_{o3} + k_{o4} < \dfrac{2}{\Omega_\varphi m_p l_p}$,

where Ω_φ is denoted as a positive constant that fulfills the following condition,

$$\left|\dot{\theta}(t) + \dot{\gamma}(t)\right| \leq \Omega_\varphi, \quad \forall t \in \mathbb{R}.$$

Then, the observer (18) converges asymptotically to the disturbance $\boldsymbol{d}$.

V. Numerical Examples

This section explores how a delay-based controller and a nonlinear disturbance observer can improve the flight performance of a tilt-rotor quadcopter. We'll first introduce the design parameters of the aircraft, controller, and observer used in this work. These specific aircraft parameters derived through software are provided in Table I.

TABLE I
Aircraft, Control and NDO parameters.

Parameter	Value	Parameter	Value
m_r	$0.5\,\mathrm{kg}$	m_p	$0.125\,\mathrm{kg}$
l_p	$0.35\,\mathrm{m}$	g	$9.81\ \frac{\mathrm{m}}{\mathrm{s}^2}$
I_r	$0.177\,\mathrm{kg{\cdot}m^2}$	I_p	$0.153\,\mathrm{kg{\cdot}m^2}$
Controller gain	**Value**	**Controller gain**	**Value** *
K_p	750	K_i	5×10^3
τ	$0.1\ s$		
Observer gain	**Value**	**Observer gain**	**Value**
ko_1	1	ko_2	1
ko_3	1.5	ko_4	2.5

To demonstrate the effectiveness of our proposed control approach, we conducted numerical simulations using a tilt-rotor quadcopter model built with commonly available, off-the-shelf components.

This scenario involves a circular path centered 6 m above the ground with a radius of 4 m. This relatively simple trajectory allows us to assess the controller's basic performance. The trajectory is described by the following parametric equation:

$$\boldsymbol{q}_d := \begin{bmatrix} h(t) \\ z(t) \\ \theta(t) \\ \gamma(t) \end{bmatrix} = \begin{bmatrix} 4\sin(\frac{\pi}{2}t) + 10 \\ 4\cos(\frac{\pi}{2}t) + 10 \\ \frac{\pi}{6} \\ \frac{\pi}{6}\sin(\frac{\pi}{2}t) \end{bmatrix}$$

We evaluated the time-delay-based controller with the nonlinear disturbance observer. Fig. 4 illustrate the resulting smooth and stable trajectory tracking, enabling precise control even in complex maneuvers. This improved performance suggests the effectiveness of the delay-based approach paired with the NDO in mitigating disturbances and achieving accurate flight paths. A closer look reveals key characteristics that help us understand specific aspects such as tracking accuracy and control efforts in all states. This will provide a nuanced understanding of the controller's strengths and weaknesses.

To perform a detailed analysis, let us focus on the states of the trajectory (Fig. 5). All measured states converge well to the desired trajectory. However, the rotational states exhibit minimal deviations caused by perturbations.

The time-delay controller exhibits a slight overdamping effect, particularly at the start of the path and during disturbances. However, it's important to note that these errors are relatively small, remaining below 2 cm and 0.5 rad, and converge to zero over time. This demonstrates the controller's effectiveness in reducing errors and achieving

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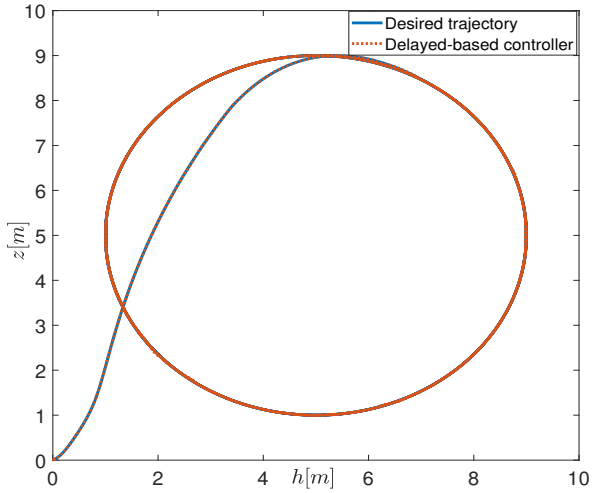

Fig. 4. Tracking of the aircraft's circular trajectory.

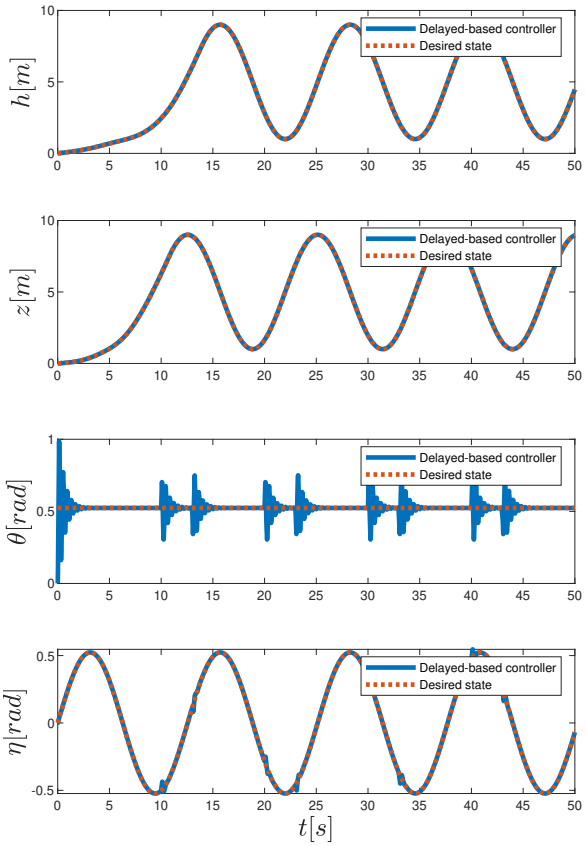

Fig. 5. Aircraft states when following a circular trajectory.

accurate trajectory tracking. Analyzing the rotor thrust (Fig. 6) is important to ensure it remains within the actuators' physical limits and prevents damage. The control strategy achieves this by generating control inputs that require a satisfactory level of effort within a short timeframe. This efficient use of control keeps the rotor thrust well within safe operating zones, enabling smooth and stable flight throughout the trajectory tracking. To simulate lifting and dropping a package, we introduced a square-wave disturbance representing the external force. The following Fig. 7

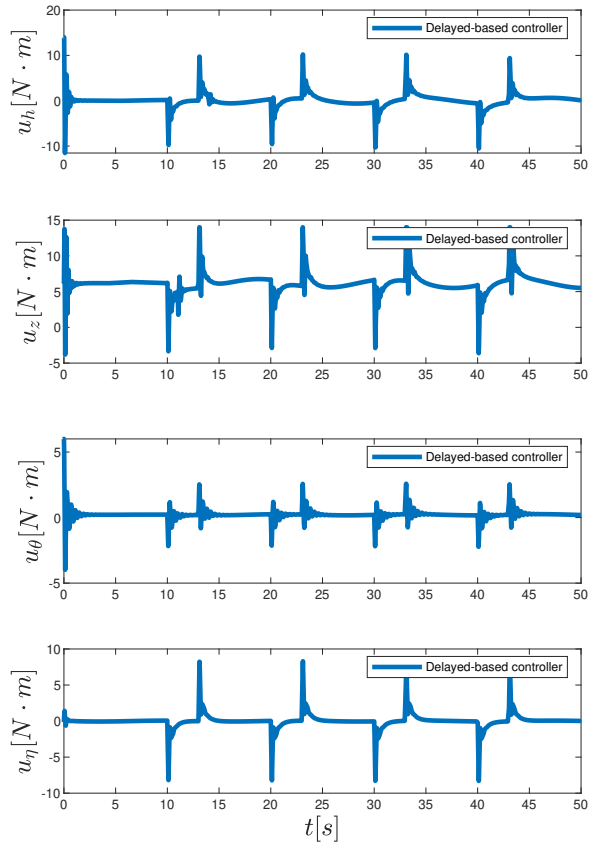

Fig. 6. Control inputs of the aircraft when following a circular trajectory.

analyze the resulting disturbances, their observations using the time-delay controller, and the observer's performance. The observer effectively estimates these disturbances, providing reliable feedback for the control system. However, it's important to acknowledge that the NDO's performance exhibits limitations during rapid changes in the external load (cargo appearance/disappearance) and at the trajectory start. These limitations are likely due to the transient nature of these events. Despite these limitations, the observer's overall performance remains satisfactory and has minimal impact on the validation of its effectiveness in disturbance estimation.

VI. CONCLUSIONS

This research explores a unique flight configuration for interactive quadcopters with tilting rotors. We developed a precise mathematical model using the Euler-Lagrange method, accurately reflecting the quadcopter's behavior.

To achieve highly accurate trajectory following and effective disturbance resistance, we propose a novel PδI controller. This controller offers greater flexibility for fine-tuning the control system, making it a strong choice for various quadcopter applications. Additionally, a modified Nonlinear Disturbance Observer (NDO) is introduced to enhance the quadcopter's performance during cargo transportation tasks.

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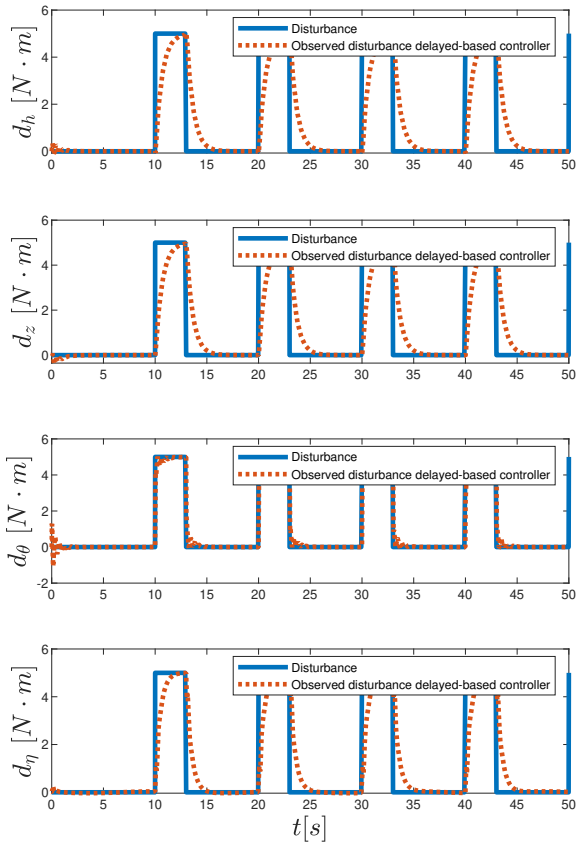

Fig. 7. Applied disturbances to the aircraft when following a circular trajectory.

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